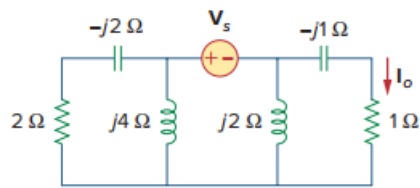


Problem

In the circuit of Fig. 9.61, find V_s if $I_o = 30\angle 0^\circ$ A.

Figure 9.61:



Step-by-step solution

Step 1 of 3

Refer to the Figure 9.61 in the textbook.

Label the nodes as shown in Figure 1.

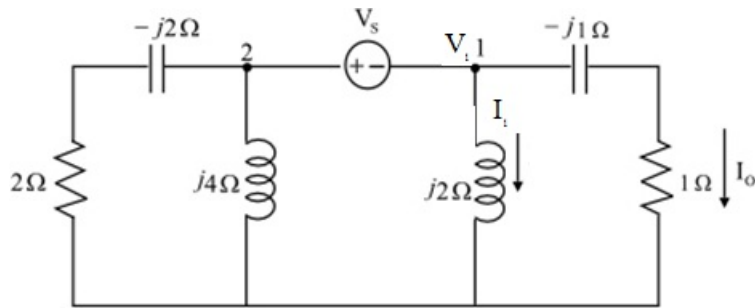


Figure 1

Step 2 of 3

The value of current is $30 \angle 0^\circ \text{ A}$.

Calculate the potential V_1 at node 1.

$$\begin{aligned} V_1 &= (1-j)I_o \\ &= (1-j)30 \angle 0^\circ \\ &= (1.414 \angle -45^\circ)30 \angle 0^\circ \\ &= 42.42 \angle -45^\circ \text{ V} \end{aligned}$$

Calculate the current I_1 flowing through the impedance $j2\Omega$.

$$\begin{aligned} I_1 &= \frac{V_1}{j2\Omega} \\ &= \frac{42.42 \angle -45^\circ}{j2\Omega} \\ &= \frac{42.42 \angle -45^\circ}{2 \angle 90^\circ} \\ &= 21.21 \angle -135^\circ \text{ A} \end{aligned}$$

The total current I which is flowing in the circuit, is the sum of currents I_o and I_1 ;

$$I = I_o + I_1.$$

Substitute the values of I_o and I_1 .

$$\begin{aligned} I &= 30 \angle 0^\circ + 21.21 \angle -135^\circ \\ &= 30 - 15 - j15 \\ &= 15 - j15 \text{ A} \\ &= 21.21 \angle -45^\circ \text{ A} \end{aligned}$$

Step 3 of 3

Now from the Figure 1, find the equivalent impedance Z_1 at node 2.

$$\begin{aligned} Z_1 &= \frac{j4(2-j2)}{j4+2-j2} \\ &= \frac{8+j8}{2+j2} \\ &= 4\Omega \end{aligned}$$

Apply Kirchhoff's voltage law to find the voltage V_s .

$$\begin{aligned} V_s + V_1 + IZ_1 &= 0 \\ V_s &= -(42.42 \angle -45^\circ + (21.21 \angle -45^\circ)4) \\ &= -(42.42 \angle -45^\circ + 84.84 \angle -45^\circ) \\ &= -(90 - j90) \text{ V} \\ &= -127.27 \angle -45^\circ \text{ V} \end{aligned}$$

Therefore, the value of the voltage V_s is $\boxed{-127.27 \angle -45^\circ \text{ V}}$.